Introductory Econometrics

Multivariate Regression Model

Ming Li

School of Management and Economics

The Chinese University of Hong Kong, Shenzhen



The variance of the OLS Estimators

Gauss Markov assumptions in MLR:

- MLR.1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + u$
- MLR.2: random sampling from the population
- MLR.3: no perfect collinearity in the sample
- MLR.4: $E(u|x_1, ..., x_k) = E(u) = 0$ (exogenous explanatory variables)
- MLR.5: $Var(u|x_1, ..., x_k) = Var(u) = \sigma^2$ (homoskedasticity)

The variance of the OLS Estimators

- **Theorem 3.1 (Unbiasedness of OLS)**

$$MLR.1-MLR.4 \Rightarrow E(\hat{\beta}_j) = \beta_j, \quad j = 0, 1, \ldots, k$$

Leaving a variable out when it should be included in multiple regression is called **underspecifying the model**, which leads to **omitted variable bias.**

- **Theorem 3.2 (Sampling variances of OLS slope estimators)**

$$MLR.1-MLR.5 \Rightarrow Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}, \quad j = 1, \ldots, k$$

The problem of almost linearly dependent explanatory variables is called **multicollinearity** (i.e. $R_j \to 1$ for some j)

The variance of the OLS Estimators

- Estimating the error variance

$$\hat{\sigma}^2 = \left(\sum_{i=1}^{n} \hat{u}_i^2 \right) / [n - k - 1]$$

An unbiased estimate of the error variance can be obtained by substracting the number of estimated regression coefficients from the number of observations. The number of observations minus the number of estimated parameters is also called the degrees of freedom. The n estimated squared residuals in the sum are not completely independent but related through the k+1 equations that define the first order conditions of the minimization problem.

$$MLR.1 - MLR.5 \quad \Rightarrow \quad E(\hat{\sigma}^2) = \sigma^2$$

- Theorem 3.3 (Unbiased estimator of the error variance)

The variance of the OLS Estimators

- Estimation of the sampling variances of the OLS estimators

The true sampling variation of the estimated β_j

$$sd(\hat{\beta}_j) = \sqrt{Var(\hat{\beta}_j)} = \sqrt{\sigma^2 / \left[SST_j(1 - R_j^2)\right]}$$

Plug in $\hat{\sigma}^2$ for the unknown σ^2

The estimated sampling variation of the estimated β_j

$$se(\hat{\beta}_j) = \sqrt{\widehat{Var}(\hat{\beta}_j)} = \sqrt{\hat{\sigma}^2 / \left[SST_j(1 - R_j^2)\right]}$$

- Note that these formulas are only valid under assumptions MLR.1-MLR.5 (in particular, there has to be homoskedasticity)

The variance of the OLS Estimators

- **How to find** $SST, SSE, SSR,$ and $\hat{\sigma}$ **in Stata?**

  Example: wage equation

  Data: WAGE1.dta

$$lwage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u$$

```
. reg lwage educ exper tenure
```

Source	SS	df	MS
Model	46.8741776	3	15.6247259
Residual	101.455574	522	.194359337
Total	148.329751	525	.28253286

Number of obs	= 526
F(3, 522)	= 80.39
Prob > F	= 0.0000
R-squared	= 0.3160
Adj R-squared	= 0.3121
Root MSE	= .44086

lwage	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
educ	.092029	.0073299	12.56	0.000	.0776292	.1064288
exper	.0041211	.0017233	2.39	0.017	.0007357	.0075065
tenure	.0220672	.0030936	7.13	0.000	.0159897	.0281448
_cons	.2843595	.1041904	2.73	0.007	.0796756	.4890435

Sampling Distributions of the OLS Estimators

- **Hypotheses testing about the β_j**

  This means we hypothesize that a population parameter equals a certain value, then use the data to determine whether the hypothesis is likely to be false.

Example: Wage equation

Natural logarithm of hourly wage

Years of formal education

Years of work-force experience

Years working in the firm

$$lwage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u$$

$$H_0 : \beta_2 = 0$$



Once we control for education and time on the current job (tenure), total workforce experience has no affect on $lwage$.

Sampling Distributions of the OLS Estimators

- **Sampling distributions of** $\hat{\beta}_j$

$$\hat{\beta}_j = \beta_j + \sum_{i=1}^{n} w_{ij} u_i$$

where the w_{ij} are functions of $\{(x_{i1}, \ldots, x_{ik}) : i = 1, \ldots, n\}$.

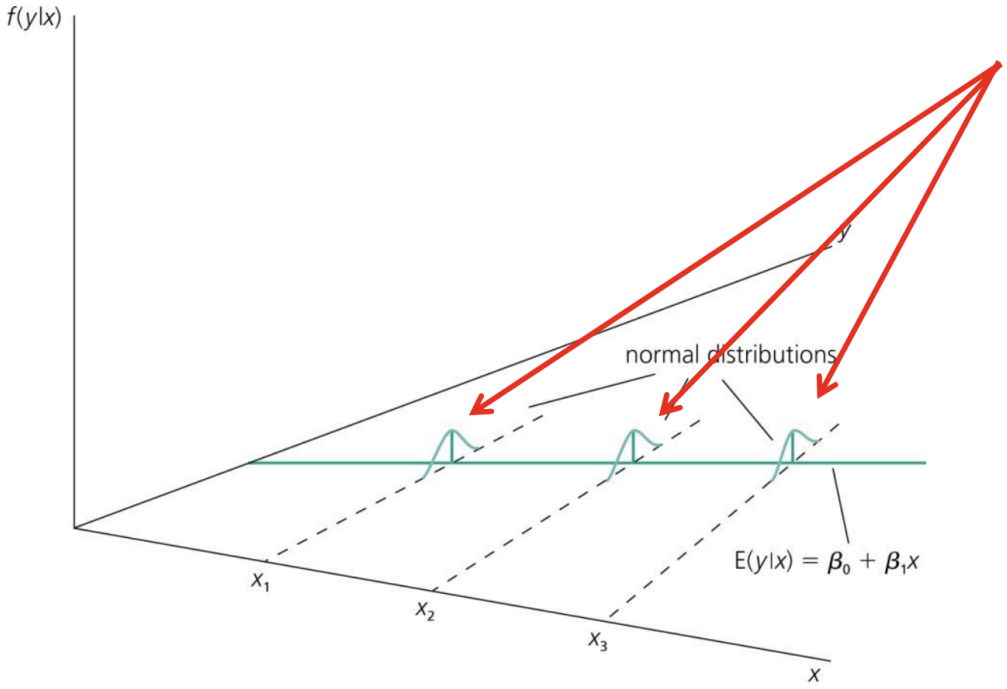
  - MLR.1 to MLR.4: $E(\hat{\beta}_j) = \beta_j$
  - MLR.1 to MLR.5: $Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$

- Hypothesis testing relies on the entire **sampling distributions** of the $\hat{\beta}_j$ Even under MLR.1 through MLR.5, the sample distributions can be virtually anything.

Sampling Distributions of the OLS Estimators

- **Assumption MLR.6 (Normality of error terms)**

$$u_i \sim N(0, \sigma^2) \quad \text{independently of} \quad x_{i1}, x_{i2}, \ldots, x_{ik}$$



It is assumed that the unobserved factors are normally distributed around the population regression function.

The form and the variance of the distribution does not depend on any of the explanatory variables.

It follows that:

$$y|\mathbf{x} \sim N(\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k, \sigma^2)$$

Sampling Distributions of the OLS Estimators

- **Discussion of the normality assumption**
  - The error term is the sum of "many" different unobserved factors. Sums of independent factors are normally distributed (CLT)
  - The normality of the error term is an empirical question
  - Examples where normality cannot hold:
    - Wages (nonnegative; minimum wage)
    - Unemployment (indicator variable, takes on only 1 or 0)
  - In some cases, normality can be achieved through transformations of the dependent variable (e.g. use log(wage) instead of wage)
  - At least the error distribution should be "close" to normal. In many cases, normality is questionable or impossible by definition

Sampling Distributions of the OLS Estimators

- **Classical linear model (CLM) assumptions**

$\underbrace{MLR.1 - MLR.5}$ "Gauss-Markov assumptions"

$\underbrace{MLR.1 - MLR.6}$ "Classical linear model (CLM) assumptions"

- u_i are independent and identically distributed (iid) as $Normal(0, \sigma^2)$

$$\hat{\beta}_j = \beta_j + \sum_{i=1}^{n} w_{ij} u_i \sim Normal[\beta_j, Var(\hat{\beta}_j)]$$

The estimators are normally distributed around the true parameters with the variance that was derived earlier

where

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

Sampling Distributions of the OLS Estimators

- **Theorem 4.1 (Normal Sampling Distributions)**

  Under the CLM Assumptions (and conditional on the sample outcomes of the explanatory variables),

$$\hat{\beta}_j \sim Normal[\beta_j, Var(\hat{\beta}_j)]$$

  and so

$$\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim Normal(0,1)$$

  The standardized estimators follow a standard normal distribution

  to test hypotheses about β_j: $sd(\hat{\beta}_j)$ depends on $\sigma = sd(u)$, which is unknown. But we have $\hat{\sigma}$ as an estimator of σ.

Testing Hypotheses About a Single Population Parameter

- **Theorem 4.1 (Normal Sampling Distributions)**

  Under the CLM Assumptions (and conditional on the sample outcomes of the explanatory variables),

$$\hat{\beta}_j \sim Normal[\beta_j, Var(\hat{\beta}_j)]$$

  and so

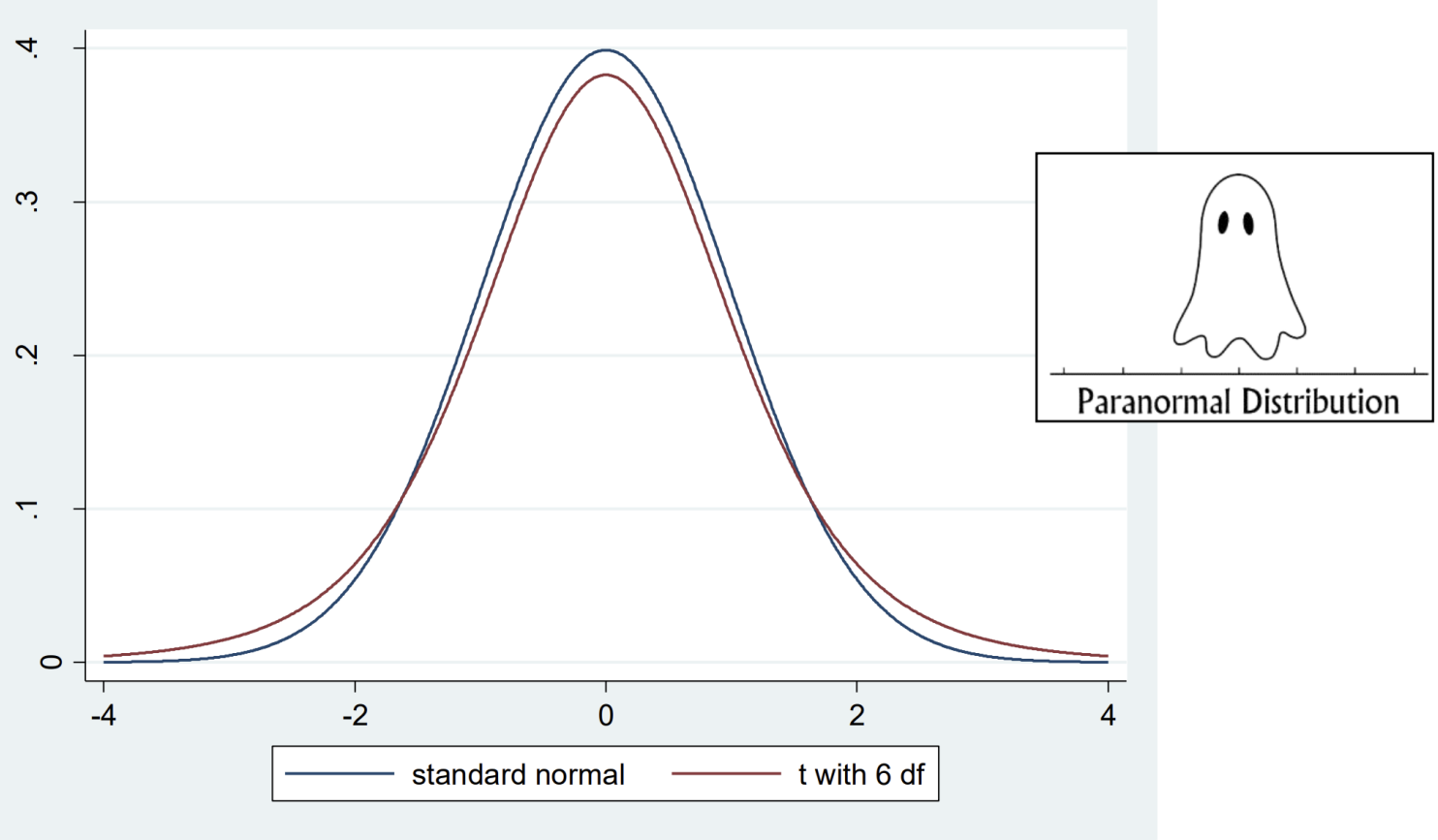
$$\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim Normal(0,1)$$

  The standardized estimators follow a standard normal distribution

  to test hypotheses about β_j: $sd(\hat{\beta}_j)$ depends on $\sigma = sd(u)$, which is unknown. But we have $\hat{\sigma}$ as an estimator of σ .

Testing Hypotheses About a Single Population Parameter

- **A standard normal distribution and a t-distribution with 6 *df*.**



Testing Hypotheses About a Single Population Parameter

- **Null hypothesis**

$$H_0 : \beta_j = 0$$

The population parameter is equal to zero, i.e. after controlling for the other independent variables, there is no effect of x_j on y

- **t-statistic (or t-ratio)**

The t-statistic will be used to test the null hypothesis.

$$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$$

The farther the estimated coefficient is away from zero, the less likely it is that the null hypothesis holds true. But what does "far" away from zero mean?

The t-statistic measures how many estimated standard deviations the estimated coefficient is away from zero.

- **Distribution of the t-statistic <u>if the null hypothesis is true</u>**

$$t_{\hat{\beta}_j} = \hat{\beta}_j / se(\hat{\beta}_j) = (\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j) \sim t_{n-k-1}$$

Testing Hypotheses About a Single Population Parameter

- **<u>Goal</u>: Define a rejection rule so that, if it is true, H_0 is rejected only with a small probability (= significance level, e.g. 5%)**

- **Testing against one-sided alternatives (greater than zero)**

  Traditional approach to hypothesis testing:

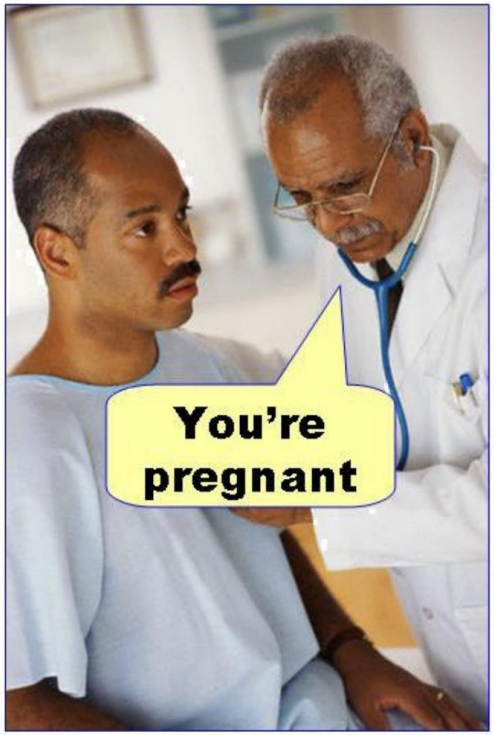
  - Choose a null hypothesis: $H_0 : \beta_j = 0$
  - Choose an alternative hypothesis: $H_1 : \beta_j > 0$
  - Choose a **significance level** (or simply **level**, or **size**) for the test. That is, the probability of rejecting the null hypothesis when it is in fact true (**Type I Error**) .
  - Calculate t-statistics $t_{\hat{\beta}_j}$
  - Choose a critical value, $c > 0$, so that the **rejection rule** is

$$t_{\hat{\beta}_j} > c$$

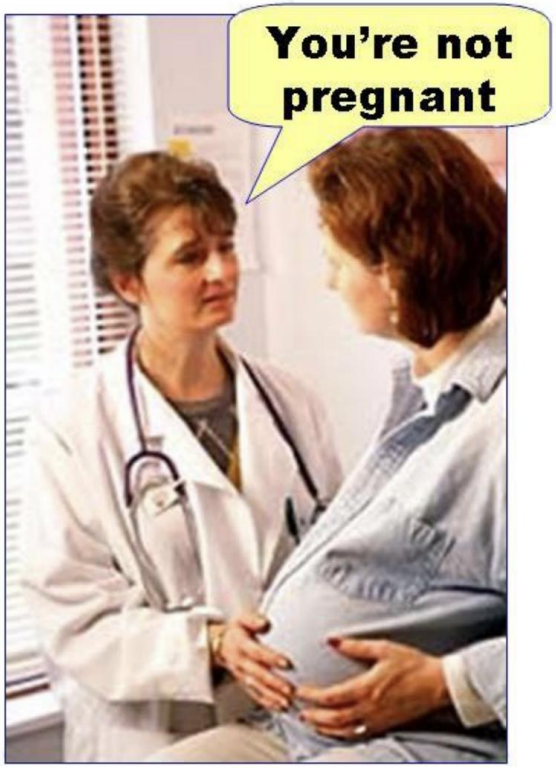
Testing Hypotheses About a Single Population Parameter

- **Type II Error**: Failing to reject the null hypothesis when it is actually false

Type I error
(false positive)

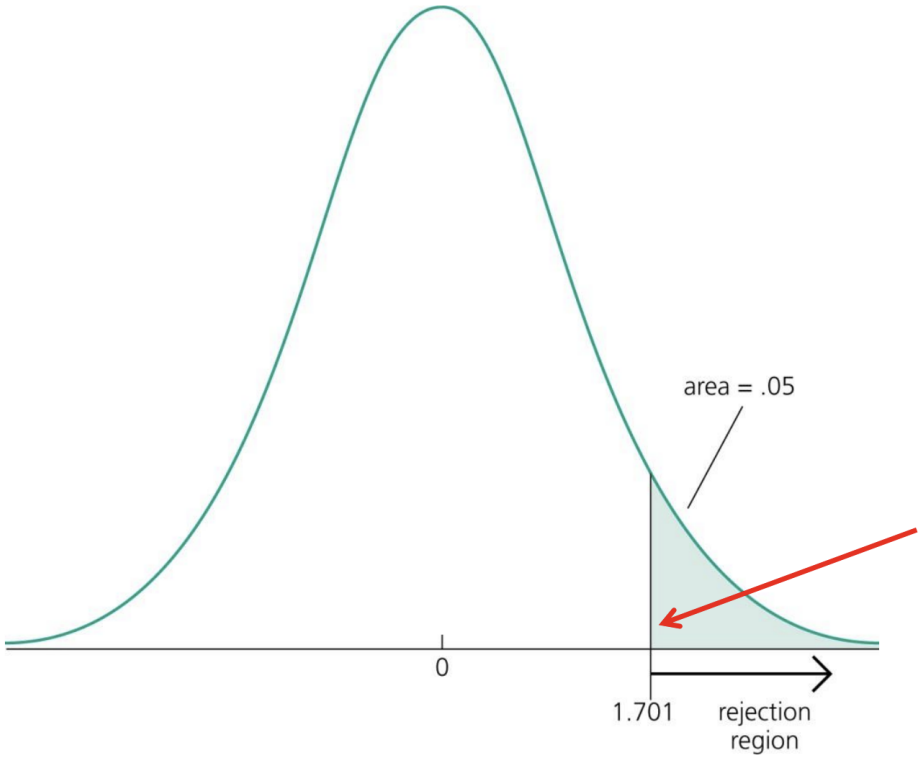


Type II error
(false negative)



Testing Hypotheses About a Single Population Parameter

- **5% rejection rule for alternative $H_1 : \beta_j > 0$ with 28 *df.***



Reject the null hypothesis in favour of the alternative hypothesis if the estimated coefficient is "too large" (i.e. larger than a critical value).

Construct the critical value so that, if the null hypothesis is true, it is rejected in, for example, 5% of the cases.

In the given example, this is the point of the t-distribution with 28 degrees of freedom that is exceeded in 5% of the cases.

Reject if t-statistic greater than 1.701

Testing Hypotheses About a Single Population Parameter

- **Example: Wage equation**

  Test whether, after controlling for education and tenure, higher work experience leads to higher hourly wages

  Data: WAGE1.dta

$$\widehat{\log}(wage) = \underset{(.104)}{.284} + \underset{(.007)}{.092}\, educ + \underset{(.0017)}{.0041}\, exper + \underset{(.003)}{.022}\, tenure$$

Standard errors

$$n = 526,\ R^2 = .316$$

Test $H_0 : \beta_{exper} = 0$ against $H_1 : \beta_{exper} > 0$.

One would either expect a positive effect of experience on hourly wage or no effect at all.

Testing Hypotheses About a Single Population Parameter

- **Example: Wage equation (cont.)**

t-statistic

$$t_{exper} = .0041/.0017 \approx 2.41$$

Degrees of freedom; here the standard normal approximation applies

$$df = n - k - 1 = 526 - 3 - 1 = 522$$

$$c_{0.05} = 1.645$$

$$c_{0.01} = 2.326$$

Critical values for the 5% and the 1% significance level (these are conventional significance levels).

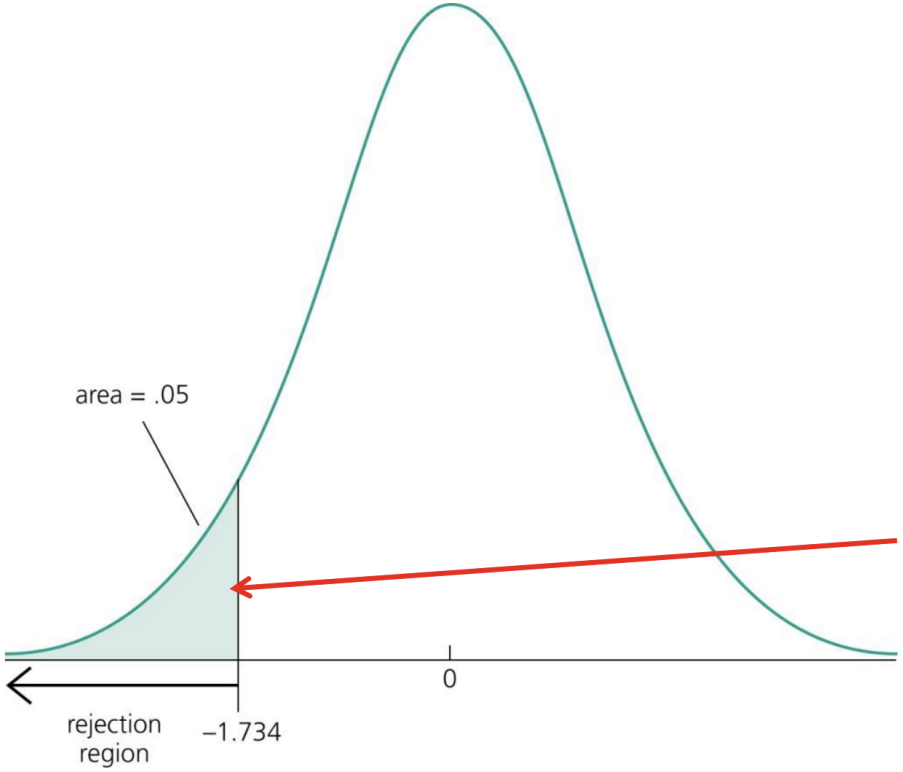
The null hypothesis is rejected because the t-statistic exceeds the critical value.

"The effect of experience on hourly wage is statistically greater than zero at the 5% (and even at the 1%) significance level."

Testing Hypotheses About a Single Population Parameter

- **Testing against one-sided alternatives (less than zero)**

  5% rejection rule for alternative $H_1 : \beta_j < 0$ with 18 *df.*



Reject the null hypothesis in favour of the alternative hypothesis if the estimated coefficient is "too small" (i.e. smaller than a critical value).

Construct the critical value so that, if the null hypothesis is true, it is rejected in, for example, 5% of the cases.

In the given example, this is the point of the t-distribution with 18 degrees of freedom so that 5% of the cases are below the point.

Reject if t-statistic less than -1.734

Testing Hypotheses About a Single Population Parameter

- **Example: Student performance and school size**

  Test whether smaller school size leads to better student performance

  Data: MEAP93.dta

Percentage of students passing maths test

Average annual teacher compensation

Staff per one thousand students

School enrollment (= school size)

$$\widehat{math10} = \underset{(6.113)}{+\ 2.274} + \underset{(.00010)}{.00046}\ totcomp + \underset{(.040)}{.048}\ staff - \underset{(.00022)}{.00020}\ enroll$$

$$n = 408,\ R^2 = .0541$$

Test $H_0 : \beta_{enroll} = 0$ against $H_1 : \beta_{enroll} < 0$.

Do larger schools hamper student performance or is there no such effect?

Testing Hypotheses About a Single Population Parameter

- **Example: Student performance and school size (cont.)**

$$t_{enroll} = -.00020/.00022 \approx -.91$$

t-statistic

$$df = n - k - 1 = 408 - 3 - 1 = 404$$

Degrees of freedom; here the standard normal approximation applies

$$c_{0.05} = -1.65$$

$$c_{0.15} = -1.04$$

Critical values for the 5% and the 15% significance level.

The null hypothesis is not rejected because the t-statistic is not smaller than the critical value.

One cannot reject the hypothesis that there is no effect of school size on student performance (not even for a lax significance level of 15%).



Testing Hypotheses About a Single Population Parameter

- **Example: Student performance and school size (cont.)**

  Alternative specification of functional form:

$$\widehat{math10} = \underset{(48.70)}{-207.66} + \underset{(4.06)}{21.16}\log(totcomp)$$

$$+ \underset{(4.19)}{3.98}\log(staff) - \underset{(0.69)}{1.29}\log(enroll)$$

$$n = 408, \quad R^2 = .0654$$

R-squared slightly higher

Test $H_0 : \beta_{\log(enroll)} = 0$ against $H_1 : \beta_{\log(enroll)} < 0$.

Testing Hypotheses About a Single Population Parameter

- **Example: Student performance and school size (cont.)**

$$t_{\log(enroll)} = -1.27/.69 \approx -1.83$$ ← t-statistic

$$c_{0.05} = -1.65$$ ← Critical value for the 5% significance level ! reject null hypothesis

The hypothesis that there is no effect of school size on student performance can be rejected in favor of the hypothesis that the effect is negative.

How large is the effect?

$$-1.29 \approx \frac{\Delta \widehat{math10}}{\Delta \log(enroll)} \approx \frac{\Delta \widehat{math10}}{100^{-1}\%\Delta enroll}$$

$$\Delta \widehat{math10} \approx -0.0129(\%\Delta enroll)$$

+ 10% enrollment ➡ -0.129 percentage points students pass test

Testing Hypotheses About a Single Population Parameter

- **Reminder about Testing**
  - Our hypotheses is about the unknown population values β_j. If in a our set of data we obtain, say, $\widehat{\beta}_j = 2.75$, we do **not** write the null hypothesis as:

$$H_0 : 2.75 = 0$$

  - Nor do we write

$$H_0 : \widehat{\beta}_j = 0$$

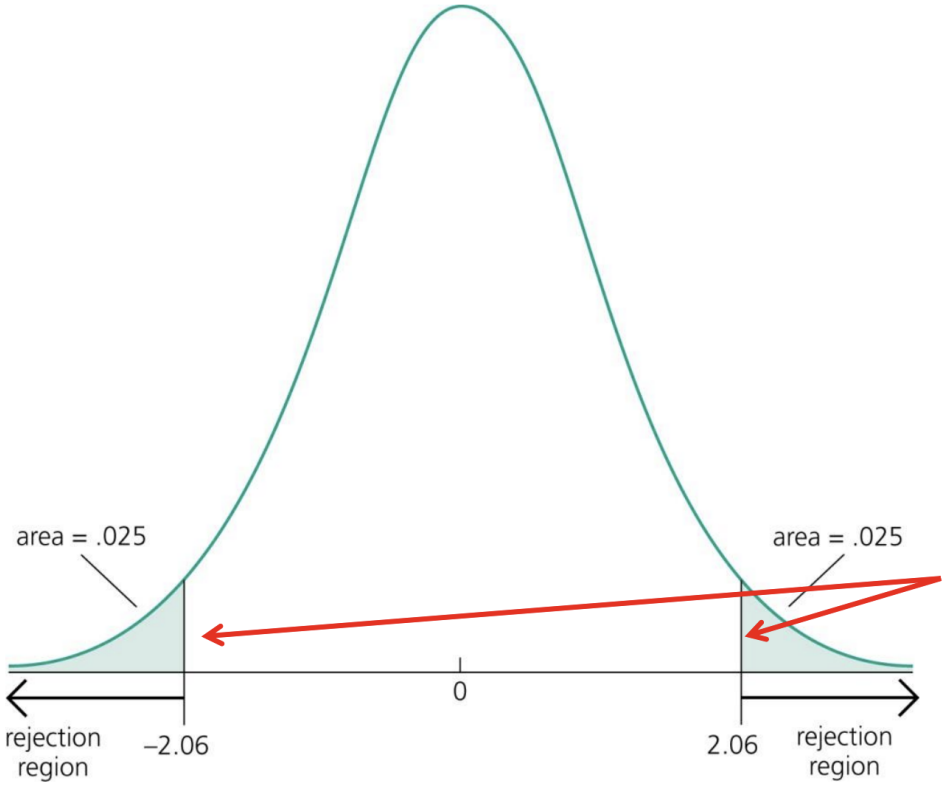
  - We do not test hypotheses about the estimate! We know what it is once we collect the sample. We hypothesize about the unknown population value:

$$H_0 : \beta_j = 0$$

Testing Hypotheses About a Single Population Parameter

- **Testing Against Two-Sided Alternatives**

  5% rejection rule for alternative $H_1 : \beta_j \neq 0$ with 25 *df.*



Reject the null hypothesis in favour of the alternative hypothesis if the absolute value of the estimated coefficient is too large.

Construct the critical value so that, if the null hypothesis is true, it is rejected in, for example, 5% of the cases.

In the given example, these are the points of the t-distribution so that 5% of the cases lie in the two tails.

Reject if absolute value of t-statistic is less than -2.06 or greater than 2.06

Testing Hypotheses About a Single Population Parameter

- **Example: Determinants of college GPA**

  Data: GPA1.dta



$$\widehat{collGPA} = \underset{(.33)}{1.39} + \underset{(.094)}{.412}\, hsGPA + \underset{(.011)}{.015}\, ACT - \underset{(.026)}{.083}\, skipped$$

$$n = 141,\ R^2 = .234$$

Lectures missed per week

$t_{hsGPA} = 4.38 > c_{0.01} = 2.58$

$t_{ACT} = 1.36 < c_{0.10} = 1.645$

$|t_{skipped}| = |-3.19| > c_{0.01} = 2.58$

The effects of hsGPA and skipped are significantly different from zero at the 1% significance level. The effect of ACT is not significantly different from zero, not even at the 10% significance level.

Testing Hypotheses About a Single Population Parameter

- **Testing more general hypotheses about a regression coefficient**
- **Null hypothesis**

$$H_0: \beta_j = a_j$$

Hypothesized value of the coefficient

- **t-statistic**

$$t = \frac{(estimate - hypothesized\ value)}{standard\ error} = \frac{(\hat{\beta}_j - a_j)}{se(\hat{\beta}_j)}$$

- **The test works exactly as before, except that the hypothesized value is substracted from the estimate when forming the statistic**

Testing Hypotheses About a Single Population Parameter

- **Example: Campus crime and enrollment**

  An interesting hypothesis is whether crime increases by one percent if enrollment is increased by one percent

  Data: campus.dta



$$\widehat{\log(crime)} = \underset{(1.03)}{-6.63} + \underset{(0.11)}{1.27}\log(enroll)$$

Estimate is different from one but is this difference statistically significant?

$$n = 97, \quad R^2 = .585$$

$$H_0 : \beta_{\log(enroll)} = 1, \quad H_1 : \beta_{\log(enroll)} \neq 1$$

$$t = (1.27 - 1)/.11 \approx 2.45 > 1.98 = c_{0.05}$$

The hypothesis is rejected at the 5% level

Testing Hypotheses About a Single Population Parameter

- **"Statistically significant" variables in a regression**
  - If a regression coefficient is different from zero in a two-sided test, the corresponding variable is said to be "statistically significant"
  - If the number of degrees of freedom is large enough so that the normal approximation applies, the following rules of thumb apply:

$|t - stat.| > 1.645 \longrightarrow$ "statistically significant at 10 % level"

$|t - stat.| > 1.96 \longrightarrow$ "statistically significant at 5 % level"

$|t - stat.| > 2.576 \longrightarrow$ "statistically significant at 1 % level"

Testing Hypotheses About a Single Population Parameter

- **<u>Guidelines for discussing economic and statistical significance</u>**
  - t testing is purely about statistical significance. It does not directly speak to the issue of whether a variable has a practically, or economically, large effect**.**
  - **Statistical significance** depends on $t_{\hat{\beta}_j}$; **Economic (practically) significance** depends on the size (and sign) of $\hat{\beta}_j$
  - If a variable is statistically significant, discuss the magnitude of the coefficient to get an idea of its economic or practical importance
  - <u>The fact that a coefficient is statistically significant does not necessarily mean it is economically or practically significant!</u>
  - If a variable is statistically and economically important but has the "wrong" sign, the regression model might be misspecified
  - If a variable is statistically insignificant at the usual levels (10%, 5%, 1%), one may think of dropping it from the regression